CHECK WHAT YOU LEARNT

Check Yourself - Unit 5

A. Choose the correct answer

1. The rain sensor works because water
   (a) is cold (b) conducts electricity (c) is heavy (d) reflects light
2. For judging how heavy the rain is, you should use
   (a) the DO pin (b) the AO pin (c) the VCC pin (d) the GND pin
3. The LDR is included in this project so that
   (a) the plate dries faster (b) the alarm stays quiet at night (c) the LEDs are brighter (d) the code is shorter
4. beep(4, 120) means
   (a) beep for 4 seconds (b) four beeps with short gaps (c) beep at 120 volts (d) wait 4 minutes
5. If your alarm behaves backwards you should
   (a) rewire everything (b) swap the > signs for < (c) buy a new sensor (d) add a resistor
6. Water pooling on the plate makes the machine report
   (a) dry for too long (b) heavy rain for too long (c) nothing at all (d) night time

B. Fill in the blanks

1 Only the ______________ of the rain sensor is allowed to get wet.

2 Two analog sensors in this project are connected to pins ______ and ______.

3 Tuning a machine to its real surroundings is called ______________.

4 Using a second piece of information to decide how to behave is called ______________ awareness.

C. Answer in one or two lines

1 Write the three readings you measured - dry, drizzle and sprayed - and the two thresholds you chose between them.

2 Explain in two sentences how water lets electricity cross the plate.

3 Why is an alarm that wakes people at 3 am a badly designed alarm?

4 Which of the three projects this year taught you the most, and why?

Extra Challenges

Finished early? Pick any of these. They all use parts you already have.

#	Challenge	Uses
1	A knob-controlled buzzer whose pitch rises as you turn it	pot + tone()
2	A rainbow fader that cycles brightness up and down on its own	for loop + PWM
3	A comfort meter that combines temperature and humidity into one score	DHT11 + maths
4	A thermostat with hysteresis that never flickers at the boundary	DHT11 + boolean
5	A humidity alarm for a book cupboard, silent below 60 percent	DHT11 + buzzer
6	A night lamp that also reports the light level every ten seconds	LDR + Serial
7	A rain gauge that counts how many minutes it rained today	rain module + millis()
8	An automatic window shutter driven by the servo	rain module + servo
9	A two-sensor weather station printing a neat table to Serial	DHT11 + LDR
10	A knob that sets the alarm threshold, so no re-uploading is needed	pot + rain module

MIXED REVISION

Twenty Questions to Test Yourself

1 What is the difference between a digital and an analog signal?
2 What value has a resistor banded red - red - brown?
3 Which pin of a module always goes to 5V?
4 What is the largest number analogRead() can return?
5 What is the largest number analogWrite() will accept?
6 Write the map() line that converts 0 to 1023 into 0 to 255.
7 Name three PWM-capable pins on the Arduino Uno.
8 Explain in one sentence how PWM dims an LED.
9 Which leg of a potentiometer goes to the analog pin?
10 What does Serial.begin(9600) do?
11 Why must the Serial Monitor's baud number match your code?
12 Which two quantities does a DHT11 measure?
13 Why must you wait two seconds between DHT11 readings?
14 What does isnan() check for?
15 What is a library, and why is it useful?
16 What is hysteresis, and what problem does it solve?
17 How does water let electricity cross the rain sensor plate?
18 Why is AO more useful than DO in the rain alarm?
19 Why does the rain alarm include a light sensor?
20 Name one thing you would add to your rain alarm with one more session.

Words to Know

Word	Meaning
Analog	A value that can be anything within a range, not just on or off.
analogRead()	Reads an analog pin and returns a number from 0 to 1023.
analogWrite()	Sends a PWM output, accepting a number from 0 to 255.
AO	The analog-out pin of a module. Gives a full reading.
Baud rate	The speed at which the board and the laptop talk. Usually 9600.
Calibration	Adjusting a machine to suit the real place it works in.
Conductivity	How easily electricity passes through a material.
Context awareness	Using extra information to decide how a machine should behave.
Decision ladder	A chain of if - else if - else, checked in order from the top.
Digital	A signal with only two values: HIGH or LOW.
DO	The digital-out pin of a module. Gives only yes or no.
float	A variable that holds a number with a decimal point.
Humidity	How much water vapour the air holds, given as a percentage.
Hysteresis	Different switch-on and switch-off points, used to stop flickering.
int	A variable that holds a whole number.
Library	Ready-made code written by someone else that you can include and use.
map()	Stretches one range of numbers neatly onto another.
Module	A small ready-made board with components already fitted.
nan	Not a number. What a sensor returns when it fails to answer.
Ohm	The unit of resistance.
Potentiometer	A resistor with a slider, used as an adjustable knob.
PWM	Pulse Width Modulation. Fast switching that makes an output look in-between.
Serial Monitor	The window where the board prints what it is thinking.
Servo	A motor that turns to an exact angle and holds it.
Threshold	The number at which a decision changes from one answer to the other.
Trimmer	A small adjustable screw that sets a module's DO threshold.
VCC	The power-in pin of a module. Connects to 5V.
Voltage divider	A circuit that splits a voltage into a smaller share.
Wiper	The middle leg of a potentiometer.

Answer Key

Section A only. Sections B and C are for you and your trainer to discuss - there is often more than one good answer.

Check Yourself - Unit 1

Q	Answer	The correct option
1	(b)	digital
2	(b)	analog
3	(b)	220 ohm
4	(b)	5V
5	(b)	a number from 0 to 1023
6	(b)	the DO threshold

Check Yourself - Unit 2

Q	Answer	The correct option
1	(b)	0 and 1023
2	(b)	0 and 255
3	(b)	convert one range to another
4	(b)	switching on and off very fast
5	(c)	D9
6	(b)	setup()

Check Yourself - Unit 3

Q	Answer	The correct option
1	(b)	wiper
2	(c)	512
3	(b)	analogWrite
4	(b)	in order, top to bottom
5	(b)	5V and GND
6	(b)	the wiper on an outer leg

Check Yourself - Unit 4

Q	Answer	The correct option
1	(b)	temperature and humidity
2	(b)	2 seconds
3	(b)	ready-made code you can use
4	(b)	the sensor failed to answer
5	(b)	percent
6	(b)	hysteresis

Check Yourself - Unit 5

Q	Answer	The correct option
1	(b)	conducts electricity
2	(b)	the AO pin
3	(b)	the alarm stays quiet at night
4	(b)	four beeps with short gaps
5	(b)	swap the > signs for <
6	(b)	heavy rain for too long

My Project Log

Fill one row after every class. This is your evidence, and it counts towards your assessment.

Session	What I built or learnt	A problem I hit	How I fixed it
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

A last word

This year your machines stopped answering yes or no and started answering how much. That is a bigger step than it sounds. Everything from a thermostat to a self-driving car is built on exactly the idea you used in these three projects: read a number, decide what it means, and act sensibly.